

3

PROBABILITY AND STATISTICS

3.1 Random Experiment

The experiment in which the outcome is uncertain is called a Random Experiment (RE).

Example: Flipping a coin, rolling a pair of dice, Picking a ball from a bag.

3.1.1 Sample Space

The set contains all the possible outcomes of a random experiment. It is denoted by 'S'.

If RE is flipping a coin, $S = \{\text{Head, Tail}\}$

If RE is rolling a dice, $S = \{1, 2, 3, 4, 5, 6\}$

3.2 Event

Any subset of sample space 'S' is called an Event.

Example: If RE is flipping a coin, then the occurring of a Head is an Event.

If RE is rolling a dice, then getting an odd number is an Event.

3.2.1 Probability of an Event

If 'A' is any event with in the sample space 'S' of a Random experiment, then the probability of event 'A' is given by

$$P(A) = \frac{\text{No. of outcomes favouring event 'A' to happen}}{\text{Total number of elements in 'S'}} = \frac{n(A)}{n(S)}$$

Probability of getting an Even Number when a dice is rolled.

$$P(\text{Even Number}) = \frac{3}{6} = 0.5$$

$$S = \{1, 2, 3, 4, 5, 6\},$$

$$A = \{2, 4, 6\}$$

Note: Probability can also be expressed as odds if favour and odds against an event:

- **Odds is favour of an event:**

Odds in **favour** of an event = Number of successes : Number of failures = m: (n – m).

- **Odds against an event:**

Odds against an event = Number of failures : Number of successes = (n – m) : m.

3.2.2 Axioms Probability

- (i) If 'A' is any event with in the sample space 'S' of a RE, then $0 \leq P(A) \leq 1$

$$\frac{0}{n(S)} \leq \frac{n(A)}{n(S)} \leq \frac{n(S)}{n(S)}$$

$$0 \leq P(A) \leq 1$$

- (ii) $P(S) = 1$

When a RE is conducted the experiment yields a possible outcome.

3.2.3 Types of Events

(i) Mutually Exclusive Events:

If A, B are two events within a sample space 'S', then A & B are said to be mutually exclusive if $A \cap B = \phi$.

Example: If 'A' is the event of getting a prime number when a dice is rolled and 'B' is the event of getting a composite number when a dice is rolled then

$$S = \{1,2,3,4,5,6\}, A = \{2,3,5\}, B = \{4,6\} \Rightarrow A \cap B = \phi \Rightarrow P(A \cap B) = 0$$

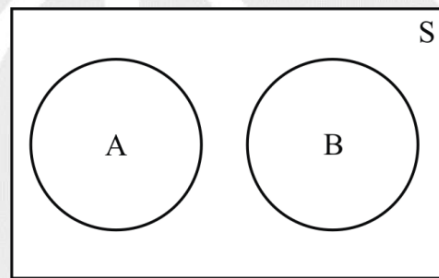


Fig. 5.1. Mutually exclusive event

(ii) Mutually Exhaustive Events:

If 'A', and 'B' are two events within a sample space 'S', then 'A' & 'B' are said to be mutually exhaustive if $A \cup B = S$

Example: If 'A' is the event of getting an odd number when a dice is rolled and 'B' is the event of getting an Even Number, then

$$A \cup B = S$$

$$S = \{1,2,3,4,5,6\}$$

$$A = \{1,3,5\}, B = \{2,4,6\}$$

$$\Rightarrow A \cup B = S$$

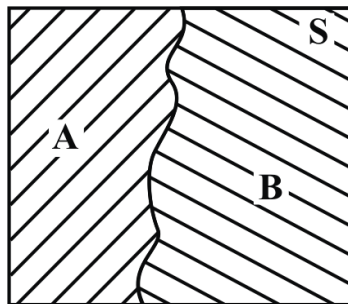


Fig. 5.2. Mutually exhaustive event

(iii) Independent Events:

Two events 'A' & 'B' within the sample space 'S' (or) within two different sample spaces 'S₁' & 'S₂' are said to be independent if $P(A \cap B) = P(A) \cdot P(B)$.

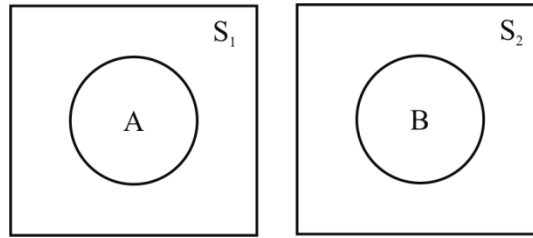


Fig. 5.3. Independent Event

(iv) Impossible Event (ϕ):

The event with zero probability is called an Impossible Event $P(\phi) = 0$.

3.3 Addition Theorem of Probability

If A, and B are two events with a sample space 'S' of a Random Experiment, then

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\frac{n(A \cup B)}{n(S)} = \frac{n(A)}{n(S)} + \frac{n(B)}{n(S)} - \frac{n(A \cap B)}{n(S)}$$

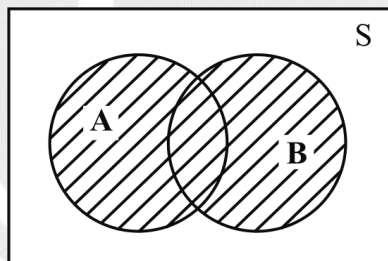


Fig. 5.4. Addition theorem

$$\Rightarrow P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

When A, and B are mutually exclusive events, $A \cap B = \phi$.

$$\Rightarrow P(A \cap B) = 0$$

$$P(A \cup B) = P(A) + P(B)$$

- If $E_1, E_2, E_3, \dots, E_n$ are mutually exclusive events ($E_i \cap E_j = \phi$), then $P(E_1 \cup E_2 \cup E_3 \cup \dots \cup E_n) = \sum_{i=1}^n P(E_i)$
 $= P(E_1) + P(E_2) + P(E_3) + \dots + P(E_n)$

3.3.1 De Morgan's Law

- $(A \cup B)^C = A^C \cap B^C$
- $(A \cap B)^C = A^C \cup B^C$

3.3.2 Union and Intersection properties

For any two events A and B:

$$(a) P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$(b) P(A^c \cap B^c) = 1 - P(A \cup B)$$

For any three events A, B and C:

$$(a) P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C)$$

$$(b) P(A^c \cap B^c \cap C^c) = 1 - P(A \cup B \cup C)$$

3.3.3 Conditional Probability:

The probability of the happening of event 'A' when it is known that event 'B' has already occurred is given by $P(A/B)$

$$P(A/B) = \frac{P(A \cap B)}{P(B)} = \frac{n(A \cap B)}{n(B)}$$

3.3.4 Joint Probability:

➤ $f(x, y)$ is the joint probability of two RV'S x, y .

➤ If the two RV are Independent then

$$f(x, y) = f(x) \cdot f(y)$$

$$➤ P(a \leq x \leq b, c \leq y \leq d) = \int_a^b \int_c^d f(x, y) dy dx$$

$$➤ f(x) = \int_{-\infty}^{\infty} f(x, y) dy$$

$$➤ f(y) = \int_{-\infty}^{\infty} f(x, y) dx$$

3.3.5 Multiplication Theorem of Probability:

If A, and B are two events within a sample space 'S', then $P(A/B) \cdot P(B) = P(B/A) \cdot P(A)$

$$P(A/B) = \frac{P(A \cap B)}{P(B)} \Rightarrow P(A \cap B) = P(A/B) \cdot P(B) \rightarrow (1)$$

$$P(B/A) = \frac{P(B \cap A)}{P(A)} \Rightarrow P(B \cap A) = P(B/A) \cdot P(A) \rightarrow (2)$$

From (1) & (2)

$$P(A/B) \cdot P(B) = P(B/A) \cdot P(A)$$

3.3.6 Total Theorem of Probability:

If $E_1, E_2, E_3, \dots, E_n$ are 'n' mutually exclusive ($E_i \cap E_j = \phi; \forall i \neq j$) and collectively exhaustive event ($E_1 \cup E_2 \cup E_3 \cup \dots \cup E_n = S$) and 'A' is any event with in the sample space 'S', then

$$P(A) = P(E_1) \cdot P(A/E_1) + P(E_2) \cdot P(A/E_2) + \dots + P(E_n) \cdot P(A/E_n)$$

$$P(A) = \sum_{i=1}^n P(E_i) \cdot P(A/E_i)$$

3.3.7 Baye's Theorem

If $E_1, E_2, E_3, \dots, E_n$ are mutually exclusive ($E_i \cap E_j = \phi \forall i \neq j$) and collectively exhaustive event ($E_1 \cup E_2 \cup E_3 \cup \dots \cup E_n = S$) and 'A' is any event with in the sample space 'S', then

$$P(E_i/A) = \frac{P(E_i) \cdot P(A/E_i)}{\sum_{i=1}^n P(E_i) \cdot P(A/E_i)}$$

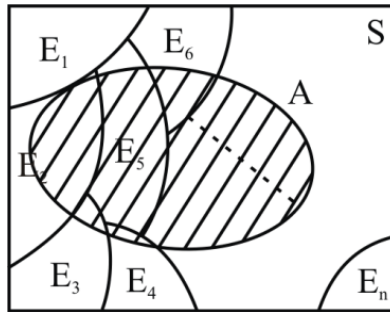


Fig. 5.5. Baye's theorem

3.3.8 Use of permutation and combination

What is combination?

A combination of 'n' objects taken 'r' at a time (r-combination of 'n' objects is an unordered selection of 'r' of the objects).

Number of ways of combining of 'r' object out of 'n' objects without repetition

$${}^nC_r = \frac{n!}{(n-r)!r!}$$

What is permutation?

A combination of 'n' objects taken 'r' at a time (r-combination of 'n' objects is an ordered selection of 'r' of the objects).

Number of ways of selection of r object out of n objects without repetition

$${}^nP_r = \frac{n!}{(n-r)!}$$

Result:

- (i) ${}^nC_r = {}^nC_{n-r}$
- (ii) ${}^nC_0 + {}^nC_1 + {}^nC_2 + \dots + {}^nC_n = 2^n$
- (iii) ${}^nC_0 + {}^nC_2 + {}^nC_4 + \dots = 2^{n-1}$
- (iv) ${}^nC_1 + {}^nC_3 + {}^nC_5 + \dots = 2^{n-1}$
- (v) $0 \cdot {}^nC_0 + 1 \cdot {}^nC_1 + 2 \cdot {}^nC_2 + \dots + n \cdot {}^nC_n = n \cdot 2^{n-1}$

Permutations with Repetition